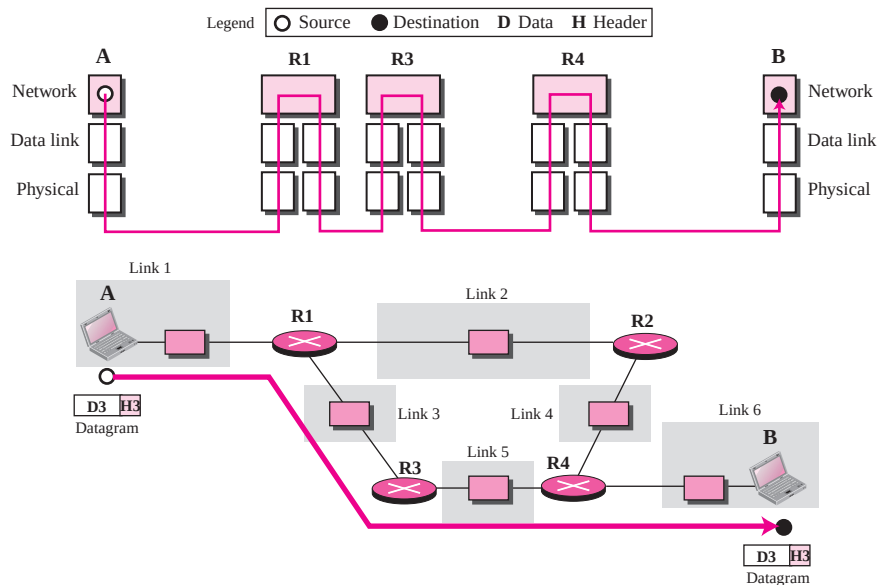


**Figure 2.12** Communication at the network layer

network layer is end to end while the communication at the other two layers are node to node. The datagram started at computer A is the one that reaches computer B. The network layers of the routers can inspect the source and destination of the packet for finding the best route, but they are not allowed to change the contents of the packet. Of course, the communication is logical, not physical. Although the network layer of computer A and B *think* that they are sending and receiving datagrams, the actual communication again is done at the physical level.

**The unit of communication at the network layer is a datagram.**

### Transport Layer

There is a main difference between the transport layer and the network layer. Although all nodes in a network need to have the network layer, only the two end computers need to have the transport layer. The network layer is responsible for sending individual datagrams from computer A to computer B; the transport layer is responsible for delivering the whole message, which is called a segment, a user datagram, or a packet, from A to B. A segment may consist of a few or tens of datagrams. The segments need to be broken into datagrams and each datagram has to be delivered to the network layer for transmission. Since the Internet defines a different route for each datagram, the datagrams may arrive out of order and may be lost. The transport layer at computer B needs to wait until all of these datagrams to arrive, assemble